Student ID _____ Student Name _____

University of the South Pacific

School of Technology Engineering Mathematics and Physics

S221 Web Applications Development

Mid Semester Test Solution

Semester 1, 2024

Possible Marks: **100**

Weight: **20%**

Time Allowed: **110** minutes

Number of Pages: **16** (including this page)

Instructions:

1. Sections A, B, C: Answer ALL questions.
2. Write ALL answers in the spaces provided in the Answer Booklet.
1. If you do need extra space for your answers, use the draft page.

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Section A: Matching Questions 20 Marks

Write the letter of the correct term that matches the given description in the space provided

A.	*<alt>*	**1.**	___C_____ Strict, Transitional and Frameset
B.	*Register.aspx.vb*	**2.**	___E_____ Indicates the code behind the page is written in Visual Basic .NET
C.	*XHTML*	**3.**	__F____ Attribute that lets you create cells that span multiple columns of table.
D.	*Browser cache*	**4.**	__G____ Protocol used during online banking.
E.	*<%@ Page Language="VB" %>*	**5.**	___A_____ Allows users to read the alternate text-based links to navigate in case images do not appear.
F.	*<colspan>*	**6.**	___I_____ Separation of presentation properties from content
G.	*HTTPS*	**7.**	____K____ Website is accessible on Chrome, IE, Safari, and Firefox
H.	*Padding*	**8.**	___J_____ Defines a table header cell.
I.	*Style declaration*	**9.**	____L____ Being able to use the web top-up feature of Digicel to successfully top up your phone
J.	*<th>*	**10.**	___D_____ Temporary storage for web graphics
K.	*Compatibility*	**11.**	____N___ Interprets HTML tags and renders the web page
L.	*Operability*	**12.**	____H____ Space between actual content and border
M.	*Multiple resolutions*	**13.**	____S____ Knowledge pertaining to networks, protocols, and Apache hardware
N.	*Rendering engine*	**14.**	___Q_____ File naming standard
O.	*Validator*	**15.**	___R_____ void using flashing content on a website as this could trigger seizures
P.	*Designing for location*	**16.**	__M_____ Website is accessible on IPhone, IPad, Android, Mac and Windows desktops

Q.	*ISO9960*	**17.**	____B____ Code behind the page and contains programming code directly related to various server controls on the form.
R.	*WCAG*	**18.**	___O_____ Checks for well-formed XHTML documents
S.	*Server administrator*	**19.**	___T_____ Reserved for domain names affiliated with the military
T.	*.mil.fj*	**20.**	____P____ Shopping cart and search functionality mostly at the top right corner of the web page

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Section B: Coding Questions 40 Marks

1) Given the following HTML content, refactor it using HTML5 semantic elements for better structure and accessibility. **(5 Marks)**

```
<div class="header">
    <h1>My Blog</h1>
    <nav>
        <ul>
            <li><a href="#">Home</a></li>
            <li><a href="#">About</a></li>
            <li><a href="#">Contact</a></li>
        </ul>
    </nav>
</div>
<div class="main">
    <div class="post">
        <h2>Blog Post Title</h2>
        <p>Blog post content...</p>
    </div>
</div>
<div class="footer">
    <p>Copyright © 2024 My Blog</p>
</div>
```

```
<header>
    <h1>My Blog</h1>
    <nav>
        <ul>
            <li><a href="#">Home</a></li>
            <li><a href="#">About</a></li>
            <li><a href="#">Contact</a></li>
        </ul>
    </nav>
</header>
<main>
    <article>
        <h2>Blog Post Title</h2>
        <p>Blog post content...</p>
    </article>
</main>
<footer>
    <p>Copyright &copy;2024 My Blog</p>
</footer>
```

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2) Given the source code below, with the help of a diagram illustrate how the web page is going to be rendered. **(10 Marks)**

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Garden of the Year Awards</title>
    <meta charset="utf-8">
    <style>
        body {
            color: #2E8B57; /* Sea Green */
            background-color: #FFFACD; /* Lemon Chiffon */
            font-family: 'Roboto', sans-serif;
        }
        main {
            background-color: #FAF0E6; /* Linen */
            padding: 5%;
            overflow: auto;
            margin-top: 1em;
            clear: both;
        }
        article {
            margin-bottom: 2%;
            background-color: #FFFFFF; /* White */
            padding: 2%;
            border: solid 0.2em #228B22; /* Forest Green */
            float: right;
            width: 28%;
            margin: 1%;
        }
        article h2 {
            text-align: center;
        }
        figure {
            text-align: center;
        }
        nav ul {
            padding: 0;
            margin: 0;
            list-style-type: none;
            background-color: #228B22; /* Forest Green */
            color: #FFFFFF; /* White */
            text-align: center;
        }
        nav li {
            display: inline-block;
            width: 30%;
        }
        nav li a {
            color: #FFFFFF;
            text-decoration: none;
            display: block;
            padding: 1em;
        }

        img { max-width: 100%; }
```

```
    </style>
</head>
<body>
    <header>
        <img src="images/banner-garden.jpg" alt="Banner for Garden of the
Year Awards">
        <h1>Garden of the Year Awards</h1>
        <p>Celebrating the best in gardening and design</p>
    </header>
    <nav>
        <ul>
            <li><a href="#">Home</a></li>
            <li><a href="#">Categories</a></li>
            <li><a href="#">Nominate</a></li>
        </ul>
    </nav>
    <main>
        <article>
            <h2>Floral Garden</h2>
            <p>Floral gardens are vibrant and full of a variety of flowers.
They represent beauty and growth.</p>
            <figure>
                <img src="images/floral.jpg" alt="Stunning floral garden">
                <figcaption>2024 Floral Garden Winner</figcaption>
            </figure>
        </article>
        <article>
            <h2>Vegetable Garden</h2>
            <p>Vegetable gardens are judged on their practicality, variety,
and sustainability.</p>
            <figure>
                <img src="images/vegetable.jpg" alt="Lush vegetable garden">
                <figcaption>2024 Vegetable Garden Winner</figcaption>
            </figure>
        </article>
        <article>
            <h2>Innovative Design</h2>
            <p>This category rewards creativity in garden design, blending
aesthetics with functionality.</p>
            <figure>
                <img src="images/innovative.jpg" alt="Garden with innovative
design">
                <figcaption>2024 Innovative Design Winner</figcaption>
            </figure>
        </article>
    </main>
</body>
</html>
```

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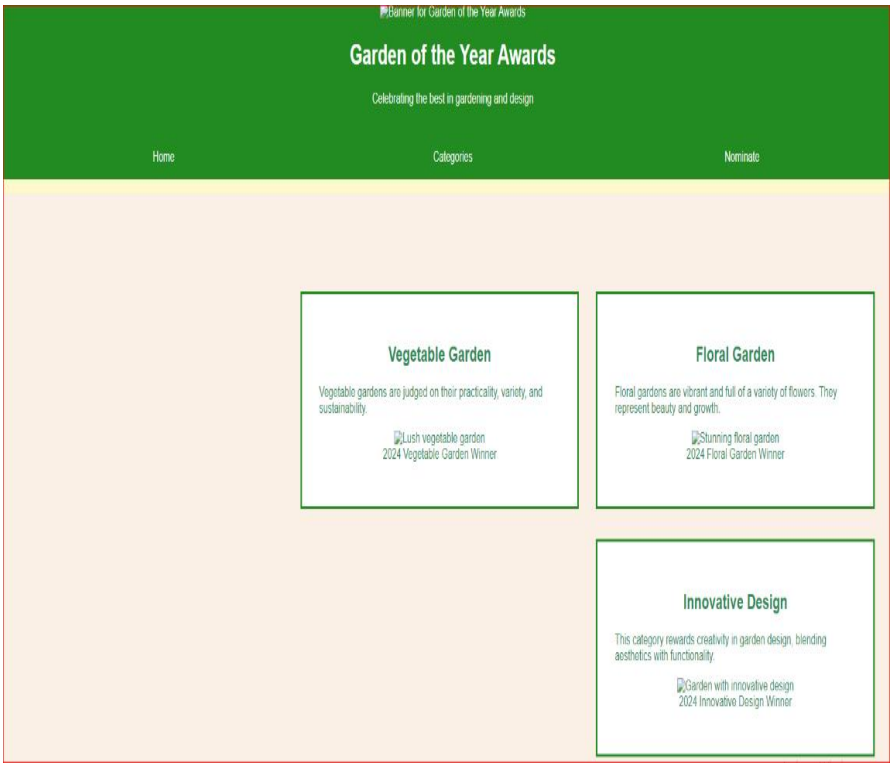

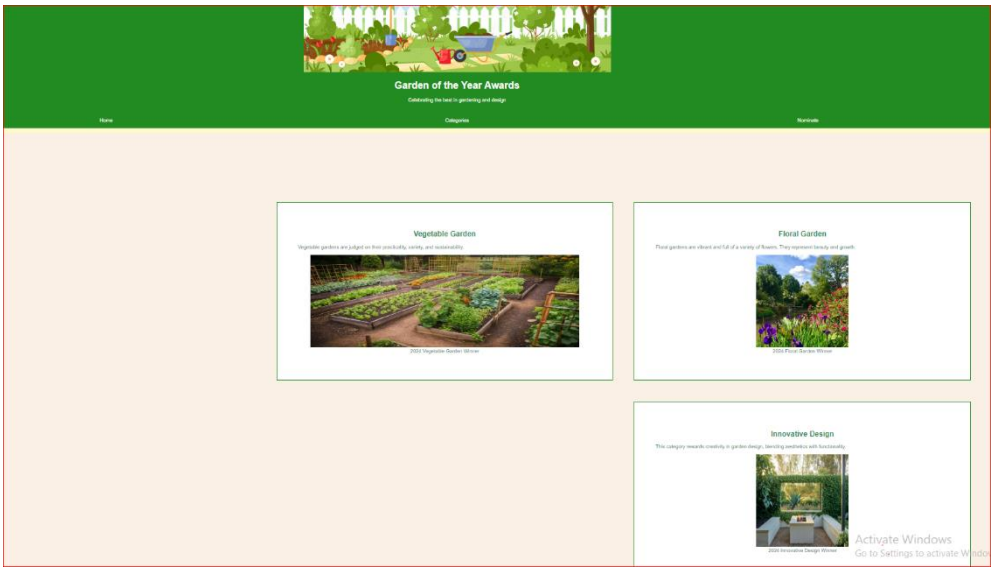

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3) Analyze the use of CSS in the Style section on the code provided in question 2 above on page 5. Style the header to have a dark background color (`#228B22; /* Forest Green */`) and the text (including the h1 title) to be white , center aligned, with a bottom padding of 1rem . **(5 Marks)**

```
header {
            padding-bottom: 1em;
            text-align: center;
            background-color: #228B22; /* Forest Green */
            color: #FFF;
        }
```

4) Given the source code below identify five specific errors, state the XHTML syntax rule that the error violates and state how to correct it using examples. **(10 Marks)**

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
    "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">
<head>

<style type="text/css">
      .style1 {
             color:#0033CC
             text-align:
             center;
                   }
</style>
<title>
  Document Title
</title>
<body>
      <h2>
            <font face="sans-serif" color="#993333" align="center">
               Introduction to XHTML &minus; Table of Contents
            </font>
      </h2>
          <ol>
            <li>
              <dl>
                <dt><b>Introduction to the
                 XHTML</b></dt>     <dd> The Three
                 variants of XHTML </dd>
              </dl>
              <dl>
```

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```
                <dt>One</dt>
                  <dd>The First XHTML Falvour Strict.</dd>

                <dt>Two</dt>
                  <dd>The Second XHTML Falvour
            Transitional.</dd>
                <dt>Three</dt>
                  <dd> The Third XHTML Falvour
            Transitional.</dd>

             </dl>
             </ol>
             </li>
</body>
</html>
```

Error 1 Missing Document Version

Syntax rule violated Document declaration includes document version, document type, and namespace

Correction Add document version <?xml version="1.0" encoding="utf-8"?

Error 2 color: #0033CC

Syntax rule violated The Cascading Style Sheet (CSS) declaration for each property should be separated with a semicolon

Correction color: #0033CC;

Error 3 <head>

Syntax rule violated XHTML will not accept ill-formed document – such as non-closing tags

Correction </head>

Error 4 use of deprecated <font> tag

Syntax rule violated XHTML: strict – does not allow deprecated tags

Correction use CSS style <h2 class="style1">

Error 5 </ol></li>

Syntax rule violated All elements should be properly nested

Correction </li></ol>

5) This question is all about ASP.NET, especially based on Lab 5 you did in IS221. Figure 1 below shows the rendered XHTML page on the client's browser with the form being filled and the user clicked the Register Me button.

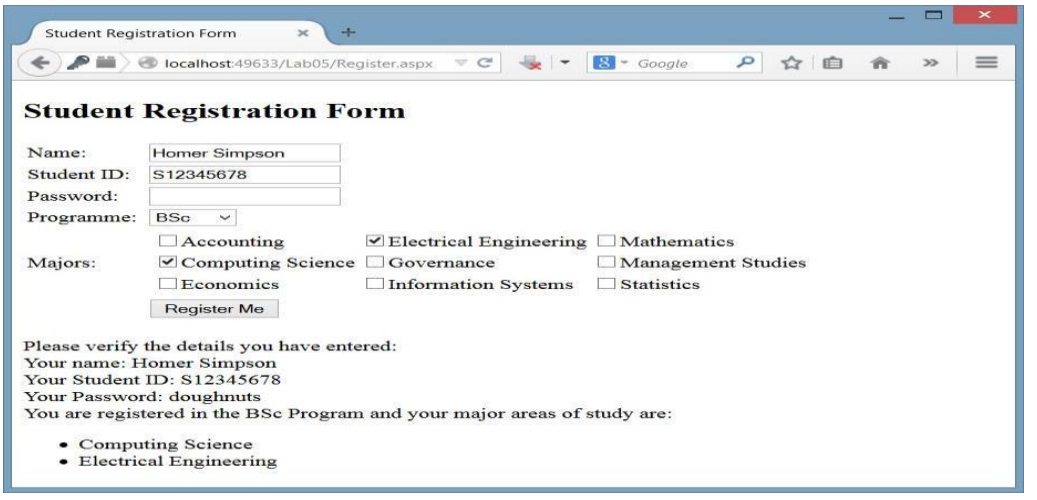

Figure 1:Register.aspx rendered successfully on the client's browser

Note that the **Register.aspx** Web form, in addition to the text content, includes a number of ASP.NET server controls or objects as summarized in the table below:

ASP.NET control/object	Name (used in coding)
3 TextBox	**txtName, txtStudentID, txtPassword**
1 DropDownList	**ddlProgram**
1 CheckBoxList	**cblMajor**
1 Button	**btnRegister**
1 Label	**lblVerify**

Use Figure 1 together with any additional information to answer the questions that follow:

a) The **general syntax** of a ASP.NET Web server control or object is given below. Every time you add a new server control, it comes with two important properties or attributes: **ID** and **Runat**. Other attributes can be added as required.

General Syntax:

```
<asp:TypeOfControl ID="ControlName" Runat="Server" />
```

Write the ASP.NET code for the **txtPassword** TextBox control. Note that you also need to add an additional attribute to the **txtPassword** TextBox control to mask the password as it is typed in by the user. **(3 Marks)**

ANSWER:

```
<asp:TextBox ID="txtPassword" runat="server" TextMode="Password"></asp:TextBox>
```

```
1 mark for the TextMode attribute
1 mark for the ID attribute
1 mark for the rest
```

b) Write the corresponding XHTML code for the **txtPassword** TextBox control. (2 Marks)

ANSWER:

```
<input name="txtPassword" type="password" id="txtPassword" />
```

```
1 mark for the type attribute.
1 mark for the rest all correct.
```

c) In the **Register.aspx** Web Form, a **DropDownList** server control or object is used. This was named as **ddlProgram**, which simply gives a list of possible programs of study a student can select from. The ASP.NET code for the **ddlProgram** DropDownList control is:

```
<asp:DropDownList ID="ddlProgram" runat="server">
        <asp:ListItem>BSc</asp:ListItem>
        <asp:ListItem>BA</asp:ListItem>
        <asp:ListItem>BCom</asp:ListItem>
        <asp:ListItem>BEng</asp:ListItem>
    </asp:DropDownList>
```

Write the corresponding XHTML code for the **ddlProgram** DropDownList server control ASP.NET code given above. **(5 Marks)**

ANSWER:

```
<select name="ddlProgram" id="ddlProgram">
    <option value="BSc">BSc</option>
    <option value="BA">BA</option>
    <option value="BCom">BCom</option>
    <option value="BEng">BEng</option>
</select>
```

```
1 mark for the select element.
1 mark for the name attribute.
```

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Section C: Short Answer Questions 40 Marks

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1) Using a diagram illustrate how you are able to view your Moodle page on your mobile phone. Ensuring to highlight the essential components of web application architecture in your illustration.

(10 Marks)

For the diagram refer to the lecture recording as it was done during our early lectures

2) Uniform Resource Identifiers, also known as Uniform Resource Locator (URL), is the name we give to Internet Addresses. It is the Webs own unique way to differentiate and identify resources on the web. The resources could be documents, images, downloadable files, services, electronic mailboxes, etc. A complete URL is made up of four parts. List and explain each part with an example?

(10 Marks)

1. Protocol

The first part of the URL "http://" refers to the Internet protocol that is to be used. In the above example the protcol that is used is HTTP.

2. Domain Name

The second part of the URL "www.elearn.usp.ac.fj" refers to server where the document will be stored. The domain is further broken down into a number of parts. The 'www' refers to the World Wide Web. The 'ac' and 'fj' are known as domain suffixes which identifies the type of website it is. In this case, 'ac' means that the server represents an academic institute and 'fj' represents a country (Fiji).

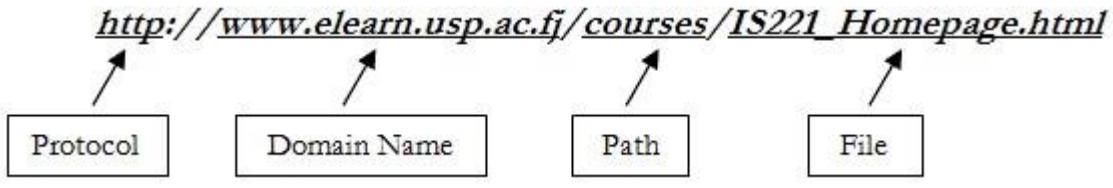
3. Path Name

The third part of the URL "courses/" represents the path name which specifies the directory on the server where the file is stored.

4. File Name

The last part of the URL "IS221_Homepage.html" is the actual file name of the file you are viewing. The filename is in two parts where you have the filename and then the file extension which determines the type of file e.g. .html, .pdf, .doc, etc...

The illustration shows the four parts of an URL.



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3) You have been contracted to design a website for an international scholarly conference. The website must be informative, easy to navigate, and accessible to a global audience. As you design this website, keep in mind the principles of web design covered in the first half of your course.

a) Explain how using tables for layout purposes can impact the design of your conference website, especially considering responsiveness and device compatibility. **(2 Marks)**

Using Tables for Layout: Tables were traditionally used for web layouts; however, they are not recommended for modern web design, particularly for responsive design. Tables restrict the fluidity of the layout, making it challenging to adapt to various screen sizes and devices. Using CSS frameworks or Flexbox/Grid systems is more appropriate for creating flexible and responsive designs that ensure content is accessible and legible on any device, which is essential for the global reach of an international conference website.

b) Describe three considerations you must take into account when designing the navigation for your website to ensure that attendees can find the information they need quickly and efficiently. **(3 Marks)**

1. **Clarity and Simplicity:** The navigation should be straightforward, using clear labels that accurately describe the sections of the website, such as "Program Schedule," "Registration," and "Keynote Speakers."
2. **Consistency:** Navigation should remain consistent throughout the entire website, so users don't get disoriented when moving from page to page.
3. **Accessibility:** Navigation should be accessible through keyboard controls and screen readers, accommodating users with disabilities and ensuring compliance with international accessibility standards, such as WCAG..

c) Highlight five principles of web accessibility that are crucial when designing your conference website to make sure it is inclusive for all attendees, regardless of ability or location.

(5 Marks)

1. **Perceivable Information**: Provide text alternatives for non-text content, create content that can be presented in different ways, and make it easier for users to see and hear content.
2. **Operable UI and Navigation**: Ensure that all users can navigate and find content, providing options to bypass blocks of content and making it operable through keyboard input.
3. **Understandable Information and UI**: Make text content readable and understandable, and ensure that web pages appear and operate in predictable ways.
4. **Robust Content**: Ensure maximum compatibility with current and future user tools, like various browsers and assistive technologies.
5. **Language and Locale**: Consider the international audience by providing options for content translation or multilingual support, and format dates, times, and currencies to accommodate international standards.

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4) Effective navigation includes providing not only links to other pages in the Web site, but also indicates to the user's location.

a) List the four navigation questions that users should be able to answer when navigating through your Web site. **(4 Marks)**

- Where am I?
- Where can I go?
- How do I get there?
- How do I get back to where I started?

b) What is information overload? **(1 Mark)**

The presence of too much information at one time on your Web pages.

c) List three ways you can limit information overload. **(3 Mark)**

Create manageable information segments.

Control page length.

Use hypertext to connect facts, relationships, and concepts.

d) The ability to use graphics for navigation is one of the most appealing aspects of the Web. To make sure your navigation graphics help rather than hinder your users, use the same graphics consistently through your Web site. Give two reasons why this is important. **(2 Mark)**

- To provide predictable navigation cues for the user
- To minimize download time